25901228 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Core Course (DSC) - MG4DSCECT204 - COMPUTER ORGANIZATION

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Very Short Answer Questions

Answer all questions

Each question carries **1** marks

1. Which type of ROM can be erased electrically and reprogrammed? [U] / [CO1]
2. Write the hexadecimal equivalent of the decimal number 17. [A] / [CO2]
3. Construct the truth table of a full adder for inputs A, B, and Cin. [A] / [CO3]
4. State the purpose of a subtractor unit. [K] / [CO1]
5. State the main purpose of using DMA. [K] / [CO2]
6. State the difference between 3-variable and 4-variable K-map. [K] / [CO3]
7. Identify the function of the data bus. [K] / [CO1]
8. Determine the output of A XNOR B for all possible inputs. [A] / [CO2]
9. State the number of select lines in a 4-to-1 multiplexer. [U] / [CO3]

10. What happens when J = K = 1 in a JK flip-flop? [U] / [CO3]

[1x10 = 10]

Part B

Short Answer Questions

Answer any **4** questions

Each question carries **5** marks

11. Discuss the advantages and disadvantages of semiconductor RAM. [U] / [CO1]

12. List the steps involved in performing binary subtraction using 2's complement. [U] / [CO2]

13. Using the truth table of a T flip-flop, determine the output sequence for the given clock input. A positive-edge–triggered T flip-flop is initially in the RESET state Q0 = 0. The given inputs are 1,0,1,1,0,1. [A] / [CO3]

14. Compare associative mapping and direct mapping [U] / [CO1]

15. What is BCD representation?Discuss when and why 6 (0110) is added during the BCD addition process. [U] / [CO2]

16. Define a parity generator. Explain the working of an even and odd parity generator with suitable examples. [K] / [CO3]

[5x4 = 20]

Part C

Essay Questions

Answer any **2** questions

Each question carries **10** marks

17. Explain different data representation types in computer organization. [K] / [CO1]

18. Simplify Boolean expressions using basic laws of Boolean Algebra and explain each step: a)A+ABA + ABA+AB b) A(A+B)A(A + B)A(A+B) [A] / [CO2]

19. Apply DeMorgan's theorem to each of the following expressions: (a) (ABC)' + (D' + E)' (b) (A + B + C)' + (D'E)' [A] / [CO3]

[10x2 = 20]